FRACTIONS (SECONDARY), (ALGEBRAIC FRACTIONS)

Name: ____________________ Date: ______________

LO: I can simplify, add, subtract, multiply and divide algebraic fractions.

What are algebraic fractions?

An **algebraic fraction** has **expressions with letters** in the numerator, denominator, or both (e.g. $\frac{x+2}{3}$ or $\frac{2x}{x-1}$).

Same rules: All fraction rules still apply. To add/subtract, find common denominator. To simplify, cancel common factors.

Key operations

●	$\frac{6x}{3} = 2x$	simplify by cancelling
●	$\frac{x}{3} + \frac{x}{4} = \frac{7x}{12}$	common denominator 12
●	$\frac{x+2}{x+2} = 1$	cancel identical factors
●	$\frac{x+2}{x} = 2$	cannot cancel part of a sum
●	$\frac{x}{3} + \frac{x}{4} = \frac{2x}{7}$	cannot add denominators

Key idea

Treat algebraic fractions like number fractions. Factorise first, then cancel common factors.

$$\frac{2x+4}{2} = \frac{2(x+2)}{2} = x + 2$$

Factorise numerator, then cancel

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simplifying

Simplify $\frac{8x}{4}$

Divide numerator and denominator by 4

$8x \div 4 = 2x;\ 4 \div 4 = 1$

$= 2x$

$\boxed{= 2x}$

Cancel the common factor of 4.

Example 2 – adding

Simplify $\frac{x}{3} + \frac{x}{4}$

Common denominator = 12

$\frac{4x}{12} + \frac{3x}{12}$

$= \frac{7x}{12}$

$\boxed{= \frac{7x}{12}}$

Multiply each fraction to get denominator 12.

Example 3 – subtracting

Simplify $\frac{2x}{5} - \frac{x}{3}$

Common denominator = 15

$\frac{6x}{15} - \frac{5x}{15}$

$= \frac{x}{15}$

$\boxed{= \frac{x}{15}}$

Multiply tops and bottoms to get a common denominator.

Example 4 – factorising to simplify

Simplify $\frac{x^2 + 3x}{x}$

Factorise numerator: $x(x + 3)$

$\frac{x(x+3)}{x}$

Cancel x: $= x + 3$

$\boxed{= x + 3}$

Factorise first, then cancel the common factor.

YOUR TURN – PRACTICE (deliberate practice)

1. Simplifying algebraic fractions
Simplify each fraction.

a) $\frac{10x}{5}$

b) $\frac{6x}{2x}$

c) $\frac{4x + 8}{4}$

d) $\frac{3x^2}{x}$

2. Adding algebraic fractions
Find a common denominator and add.

a) $\frac{x}{2} + \frac{x}{3}$

b) $\frac{x}{4} + \frac{x}{5}$

c) $\frac{2x}{3} + \frac{x}{6}$

d) $\frac{x}{2} + \frac{3}{4}$

3. Subtracting algebraic fractions
Find a common denominator and subtract.

a) $\frac{x}{2} - \frac{x}{5}$

b) $\frac{3x}{4} - \frac{x}{3}$

c) $\frac{5x}{6} - \frac{x}{2}$

d) $\frac{x+1}{3} - \frac{x}{4}$

4. Factorising to simplify
Factorise the numerator or denominator, then cancel.

a) $\frac{x^2 + 2x}{x}$

b) $\frac{3x + 6}{3}$

c) $\frac{x^2 - 4x}{x}$

d) $\frac{2x^2 + 6x}{2x}$

5. Mixed algebraic fractions
Simplify fully.

a) $\frac{12x}{4x}$ b) $\frac{x}{3} + \frac{2x}{9}$ c) $\frac{x^2 + 5x}{x}$ d) $\frac{4x}{6} - \frac{x}{3}$ e) $\frac{2(x + 1)}{x + 1}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{6x}{3} = 2x$ ☐	**b) $\frac{x+2}{x} = 2$ (cancel the x)** ☐	**c) $\frac{x}{3} + \frac{x}{4} = \frac{7x}{12}$** ☐	**d) $\frac{x}{3} + \frac{x}{4} = \frac{2x}{7}$** ☐	**e) Factorise before cancelling** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: ________________	Reason: ________________	Reason: ________________	Reason: ________________	Reason: ________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Simplify $\frac{x^2 + 5x + 6}{x + 2}$. (Hint: factorise the numerator as a quadratic first.)

Expression: ____________________________

Simplified: ____________________________

b) Show that $\frac{1}{x} + \frac{1}{x+1} = \frac{2x+1}{x(x+1)}$. Hence find the value when **x = 3**.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?
Did I check my answer using a different method?
Does my answer look reasonable?

WELL DONE!